

ISO/IEC 42001 AI System Cost-Benefit & Economic Analysis

Document Reference: CAW-EVID-CBA-2026-01
System: Case Ace v2.0 Client Consultation Recording & Note Drafting System
Organisation: Citizens Advice Wandsworth (CAW)
Standard: ISO/IEC 42001:2023 Clause B.5 (Resource Management & Value Realisation)
Effective Date: 2026-09-02
Status: Formally Verified Economic Analysis
Classification: Official / Governance

1. Scope & Objective

In accordance with ISO/IEC 42001:2023 Clause B.5 (Management of Resources for AI Systems), this document provides the formal Cost-Benefit Analysis (CBA) evaluating the financial sustainability, unit economics, and capacity return on investment for Case Ace v2.0.

2. Quantitative Financial & Operational Metrics

QUANTITATIVE COST & CAPACITY BENCHMARK		
Metric Category	Pilot Phase (6 Weeks)	Full Roll-Out (Month)
Adviser / Caseworker Headcount	15 Advisers	75 Advisers & Vols
Casenote Volume	250 Total	2,750 / Month
Total Operational Cost	**£81.66 Total**	**£217.65 / Month**
Monthly Operating Cost	**£54.44 / Month**	**£217.65 / Month**
Cost Per Casenote	**£0.33 / Note**	**£0.079 (7.9p) /
Note**		
Cost Per Adviser	**£3.63 / Adv / Month**	**£2.90 / Adv /
Month**		
Caseworker Time Saved	62.5 Hours	687.5 Hours / Month
Capacity Value Unlocked (@ £18.50/hr)	£1,156.25	£12,718.75 / Month
Return on Investment (ROI Ratio)	**14.2 : 1**	**58.4 : 1**

3. Granular Cost Center Model

```
graph LR
  subgraph VertexAI["Vertex AI Gemini 1.5"]
    V1["Flash Input: £0.06/1M tokens"]
    V2["Flash Output: £0.24/1M tokens"]
    V3["Unit Cost: £0.00062 / note"]
  end

  subgraph STT["Google Cloud Speech v2"]
    S1["Local Whisper: £0.00"]
    S2["Cloud Dual-Pass: £0.58/call"]
    S3["10% Fallback: £0.058 / note"]
  end

  subgraph Hosting["Cloud Run europe-west2"]
    H1["Compute Instances: £28.00/mo"]
    H2["Load Balancer: £22.00/mo"]
    H3["Logging/Secrets: £5.00/mo"]
  end

  VertexAI --> Total["Total Monthly: £217.65"]
  STT --> Total
  Hosting --> Total
```

4. Sensitivity & Worst-Case Risk Boundary

Zero Cloud STT Scenario (100% Local WASM): £58.15 / month (£0.021 / casenote, £0.78 / adv / mo).
Baseline Realistic Scenario (10% Cloud Fallback): £217.65 / month (£0.079 / casenote, £2.90 / adv / mo).
High Fallback Scenario (25% Cloud Fallback): £456.90 / month (£0.166 / casenote, £6.09 / adv / mo).

Upper Bound Cap (100% Cloud Transcription): £1,653.15 / month (£0.601 / casenote, £22.04 / adv / mo).

5. Formal Verification Sign-Off

Finance & Audit Committee Review: 2026-09-02 (Noted and Recommended to Board).

ISO/IEC 42001 Compliance Lead: Signed on file.

Master Financial Model Reference: docs/pilot/cost-estimate.md.